

Sasi Vaibhav Sanka

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SUMMARY

Results-driven AI & Data Science student specializing in developing and deploying scalable Computer Vision, Deep Learning, and RPA solutions. Proven success in executing production-ready ML systems (face recognition, retail automation). Demonstrated rapid problem-solving and delivery, ranking **Top 10** in VIE AI-Thon. Seeking to leverage technical depth and strong analytical skills in a challenging technology consulting environment.

EDUCATION

- BS in Data Science and Applications – IIT Madras, Chennai, India** Dec 2023 – Present
- **Coursework:** Data Structures and Algorithms (DSA), DBMS, Artificial Intelligence, Machine Learning, Data Analysis & Visualization. **Ranked #1 Course Topper.**
- B. Tech in Artificial Intelligence and Data Science – KLH University, Hyderabad, India** Jun 2023 – May 2027
- **Coursework:** Deep Learning, Advanced Algorithms, Web & App Development. | **GPA: 9.36 / 10.0** (Top 2% of cohort)

EXPERIENCE

- Machine Learning Intern – Amnet Digital, Hyderabad** May 2025 – Jul 2025
- Engineered an advanced facial recognition system achieving high accuracy (**95%+**) even under mask/partial occlusion, enhancing real-world surveillance robustness across **10+** test streams.
 - Optimized a real-time video processing pipeline to ensure **low-latency inference** (**<30ms**) for integration with live security camera feeds.
 - Developed a retail shelf-monitoring Proof-of-Concept (POC) using YOLOv8, achieving over **95%+** accuracy in detecting out-of-stock and misplaced inventory across **500+** SKU variations.
 - Prepared the complete system for deployment, validating performance on public datasets against live in-store data streams.

COMPETITIONS & HACKATHONS

- Awetales Dubsmart AI Hackathon – 1st Prize Winner** 2026
- Built an advanced automated audio dubbing system that effectively clones voice profiles across **100+** languages with high-fidelity speaker matching.
- Microsoft DevNnovate Hackathon – Participant** Sep 2025
- Developed a Computer Vision model for real-time detection of emergency space station objects by simulating scenarios.
 - Created and augmented a custom dataset and trained the model, integrating it into a real-time detection simulation within the intense **48-hour** event.
- Deploy Thon – Participant** Aug 2025
- Engineered a scalable web-scraping module and successfully developed and deployed a fully functional Lost and Found management system using web technologies, demonstrating deployment capability.

RESEARCH & PUBLICATIONS

- Age-Invariant Kinship Verification using Deep Generative Models** Sep 2025
- CycleGAN, Siamese Networks*
- Developed a novel kinship verification framework using CycleGAN for age normalization paired with triplet-loss Siamese networks, improving baseline validation marks.
 - Implemented a **3-phase** pipeline (3D reconstruction, feature extraction, classification) to achieve improved accuracy on the KinFaceW benchmark.
- Colorization of Grayscale Images** Dec 2024
- Autoencoders, U-Net*
- Implemented a deep autoencoder with U-Net skip connections for automated image colorization, validating performance using industry-standard high-fidelity PSNR and SSIM metrics.

PROJECTS

- Shelf Monitoring & Retail Automation PoC** Jul 2025
- Computer Vision, YOLOv8*
- Created a real-time inventory tracker to identify misplaced and missing items, optimized for **edge-device deployment** with low compute overhead.
- Medicare Disease Prediction System** Dec 2024
- ANN, Boosting, Django*
- Built a full-stack platform predicting multiple clinical outcomes from user symptoms using ANN; integrated features include hospital finder and appointment booking.
- Optimal Mathematical Programming Solver** Jan 2025
- Python, Django, LP Optimization*
- Designed an intuitive educational tool implementing Simplex, Graphical, and Transportation algorithms, providing step-by-step solver logic.
- Study Planner (Optimal Study Time Allocator)** Jan 2025
- Python, Django, Simplex*
- Developed an intelligent study planner that optimally allocates study hours using the Simplex method, dynamically updating based on user inputs.

TECHNICAL SKILLS

- Languages:** Python, Java
- AI/ML & DL:** PyTorch, TensorFlow, Scikit-learn, OpenCV, Ultralytics, NumPy, Pandas, CNN, GANs, Vision Transformers
- Web/App:** Django, HTML/CSS, Web Scraping
- Core CS:** Data Structures & Algorithms (DSA), Database Management Systems (DBMS), Optimization, Operating Systems (Linux)
- Tools:** Git, GitHub, Tableau, Google Colab, Jupyter

AWARDS & CERTIFICATIONS

- AWS Certified Cloud Practitioner
- Course Topper (IIT Madras BS DSA)
- NASSCOM Exploratory Data Analytics Gold
- Advanced RPA Essentials Certification
- Convolutional Neural Networks (Coursera)
- GitHub Foundations Certificate (Microsoft)
- Coding Challenge Certificate (IITM / KLH)